

มหาวิทยาลัยมหิดล
Mahidol University



ITCS215
Introduction to Interactive Multimedia Technology

Semester 1/2026

Uniform Image Resizing
Non-Uniform Image Scaling (Seam Carving)

Assoc. Prof. Dr. Worapan Kusakunniran

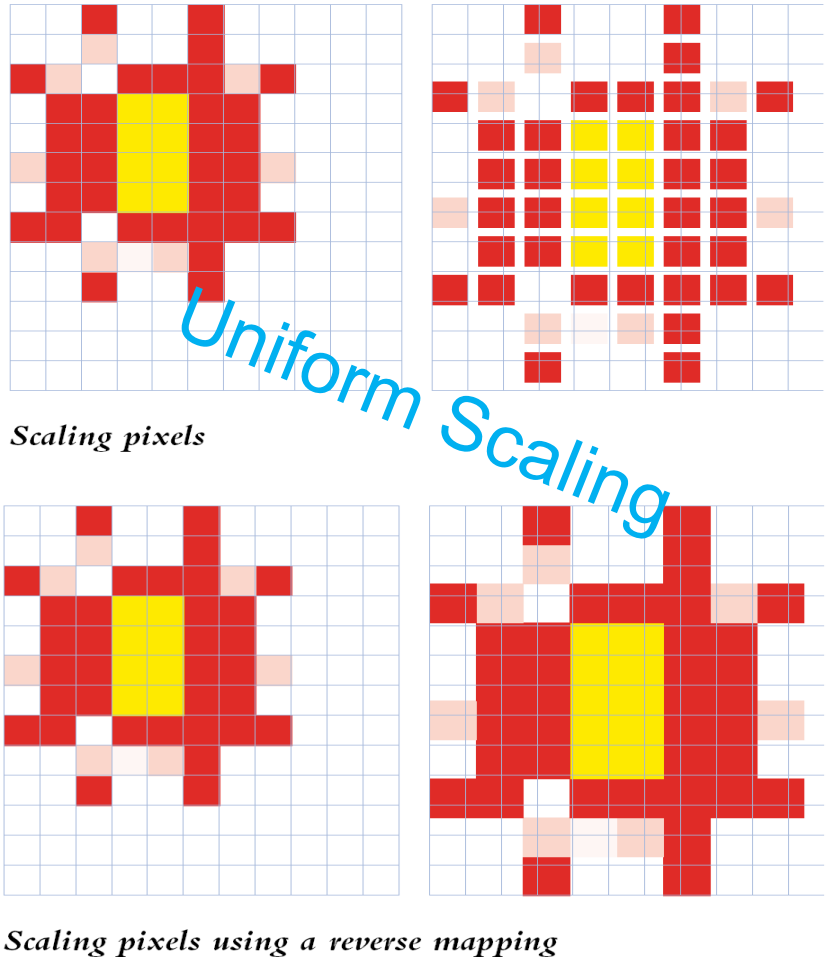
Image Resizing

- **Interpolation** is needed because of the finite size of pixels
  - Finite pixel doesn't have real decimal index
  - Interpolation can obtain approximate value to fit in index based on proportional values between it

Nearest neighbor

Bilinear interpolation

Bicubic interpolation



Limitation of Interpolation

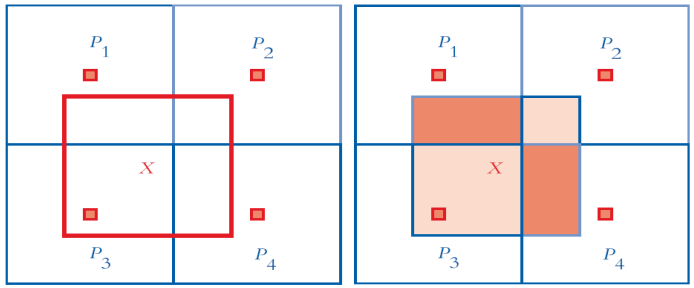


Nigel Chapman and Jenny Chapman, Digital Multimedia

Image Scaling (cont.)

See example

- Suppose that we are scaling an image
- Original image coordinate (x, y)
- Scaled image coordinate (x', y')



Pixel interpolation *Areas for bilinear interpolation*

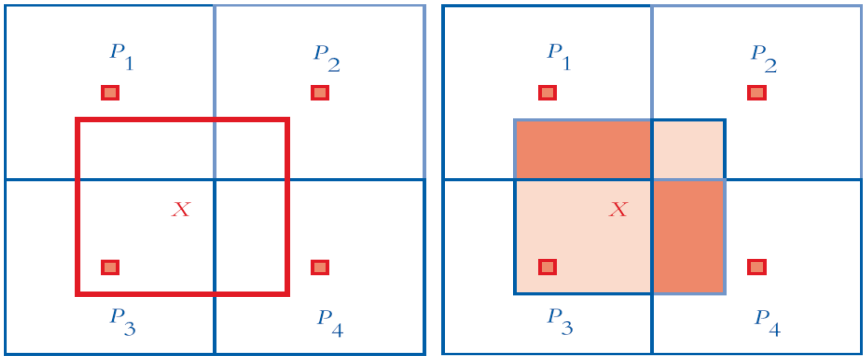
- When we want to fill-in color for each pixel at (x', y') in the scaled image, we will use color of pixel at (x, y) from the original image, where $(x, y) = (x'/s, y'/s)$, s is scaling factor
- x and y are often not integer, thus such pixel does not actually exist
- (x, y) are shown in the above figure as x, surrounding by 4 pixels P_1, P_2, P_3, P_4

Interpolation

3 types of interpolations

- Nearest neighbor
- Bilinear interpolation
- Bicubic interpolation

Interpolation: Nearest Neighbor



Pixel interpolation

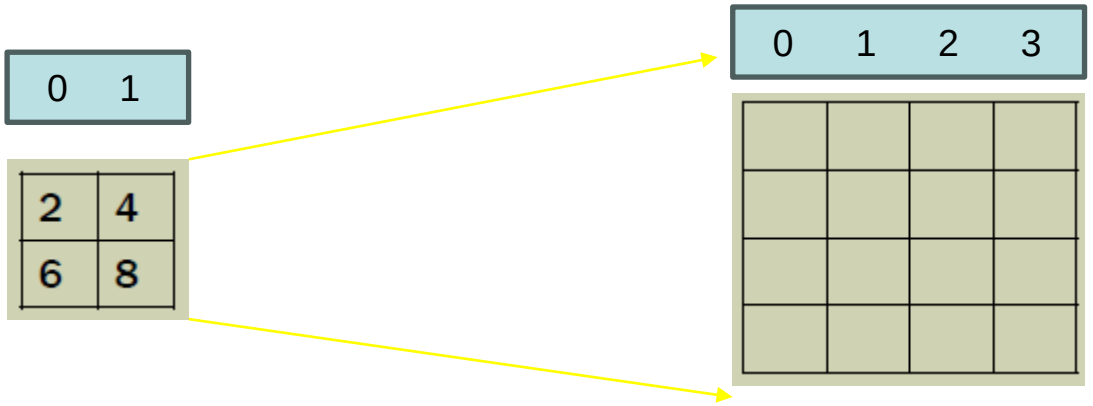
Areas for bilinear interpolation

See example

- Nearest neighbor
  - Use value of pixel whose center is closest in the original image to real coordinates of ideal interpolated pixel (proportional round up)
  - In this case is P3

Class Activity 2.1

Nearest Neighbor



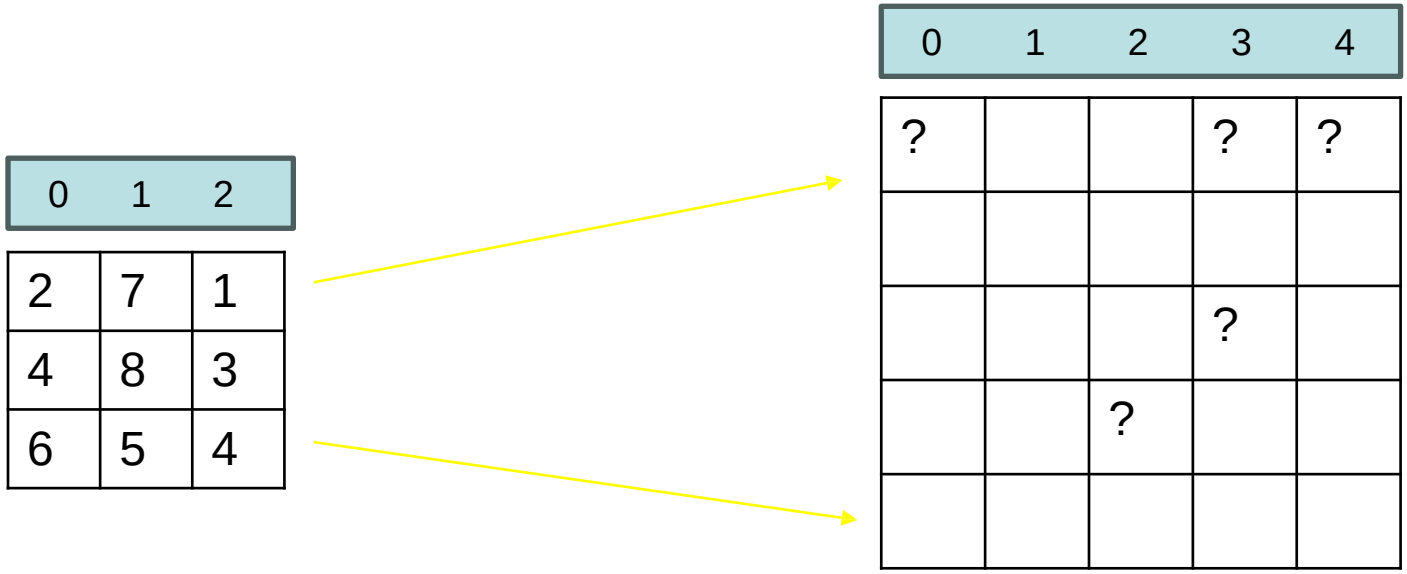
<= 0.5
Scale down

- Calculate the values in the new image
- Show the details of your solution and calculation

$(x,y) = (x'/s, y'/s)$, **s is scaling factor**
S_w = Width_out/Width_in
S_h = Height_out/Height_in

Class Activity 2.2

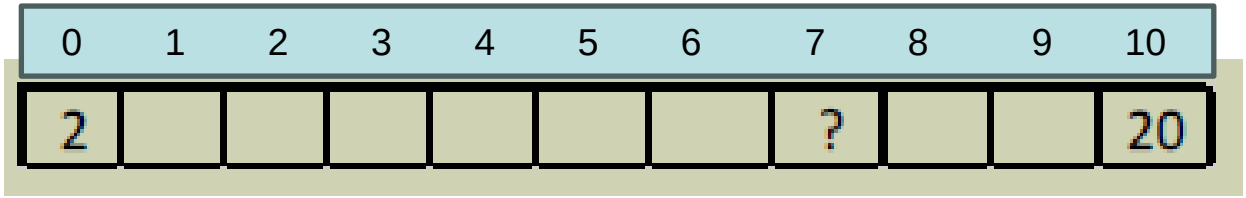
Nearest Neighbor



- Calculate the values in the ? by showing the details of your solution and calculation

Interpolation: Linear Interpolation

- Before talking about ***Bilinear (2D)***, let's see the ***Linear (1D)*** interpolation



See explanation

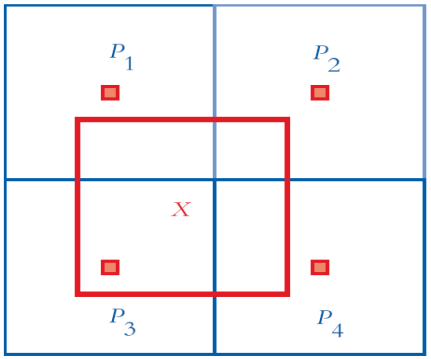
Interpolation: Bilinear Interpolation

- Bilinear interpolation
  - Use value of all four adjacent pixels, weighted by intersection with target pixel

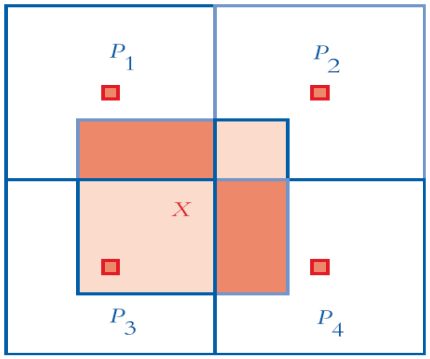
- If $P1(x1, y1), P2(x2, y1), P3(x1, y2), P4(x2, y2)$ and $P(x, y)$, thus the value of pixel (x', y') is

$$(x2 - x)(y2 - y)P_1 + (x - x1)(y2 - y)P_2 + (x2 - x)(y - y1)P_3 + (x - x1)(y - y1)P_4$$

See example



Pixel interpolation



Areas for bilinear interpolation

Interpolation: Bicubic Interpolation

- Use values of all four adjacent pixels, weighted using **cubic splines**

Nearest-neighbor interpolation is quickest but produces poor-quality results Bicubicis slowest but produces very good results



Nearest–neighbour (left), bilinear (middle) and bicubic (right) interpolation

:tween

Interpolation Results

Nigel Chapman and Jenny Chapman, Digital Multimedia

Interpolation Results





Bilinear

Nearest Neighbor

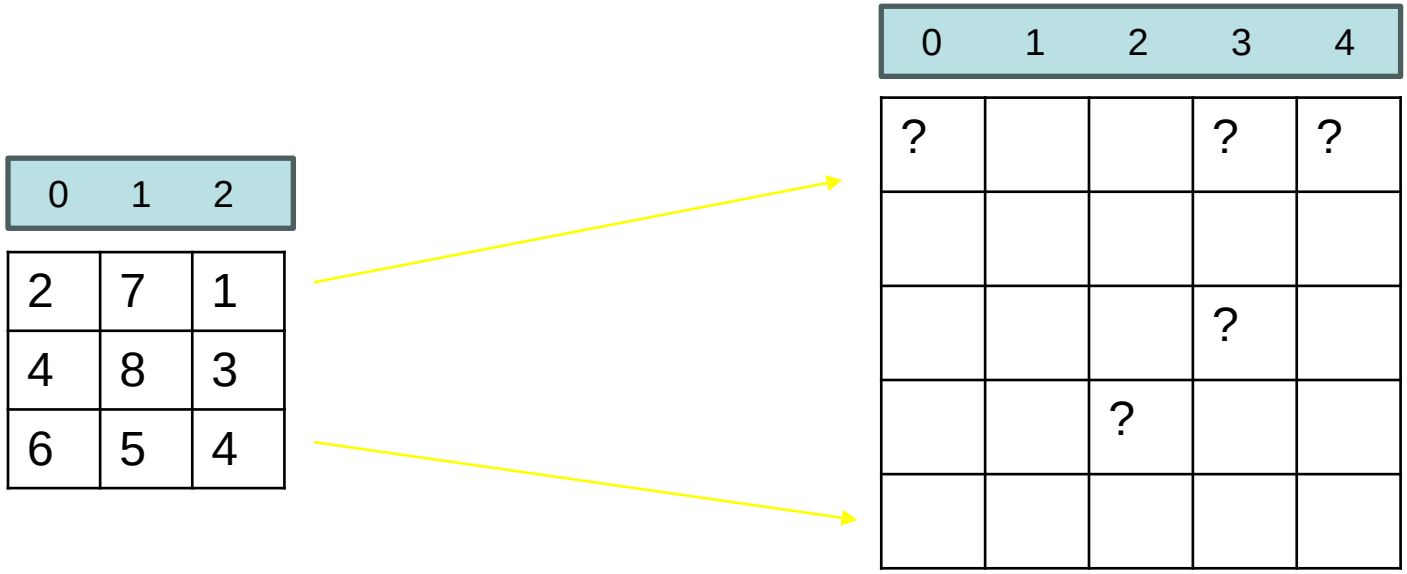




Bicubic

Class Activity 2.3

Bilinear Interpolation



- Calculate the values in the ? by showing the details of your solution and calculation

$$(x2-x)(y2-y)P_1+(x-x1)(y2-y)P_2+(x2-x)(y-y1)P_3+(x-x1)(y-y1)P_4$$

Class Activity 2.4
(Python)

Image Resize

- Use your own picture
- Increase and decrease size for 4 times
- Write a NOTE in comments for

  - Compare between two interpolation methods, which one is better and why

  - Compare between increase and decrease a size, which one has a higher chance to distort an image quality and why

More Intelligent Image Resizing

- Content Aware Image Resizing (Seam Carving)

Non-Uniform Scaling

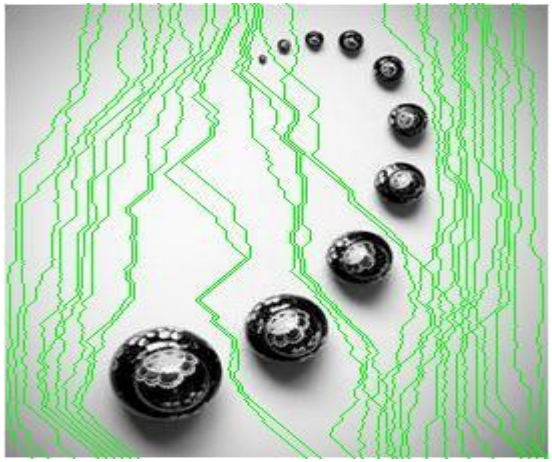


Seam Carving

- Seam Carving
  - Consider image content by carving out the **least important seam**
  - Seam - is a connected path of **low energy pixels** in an image



Original Image



Original Image with Vertical Seam

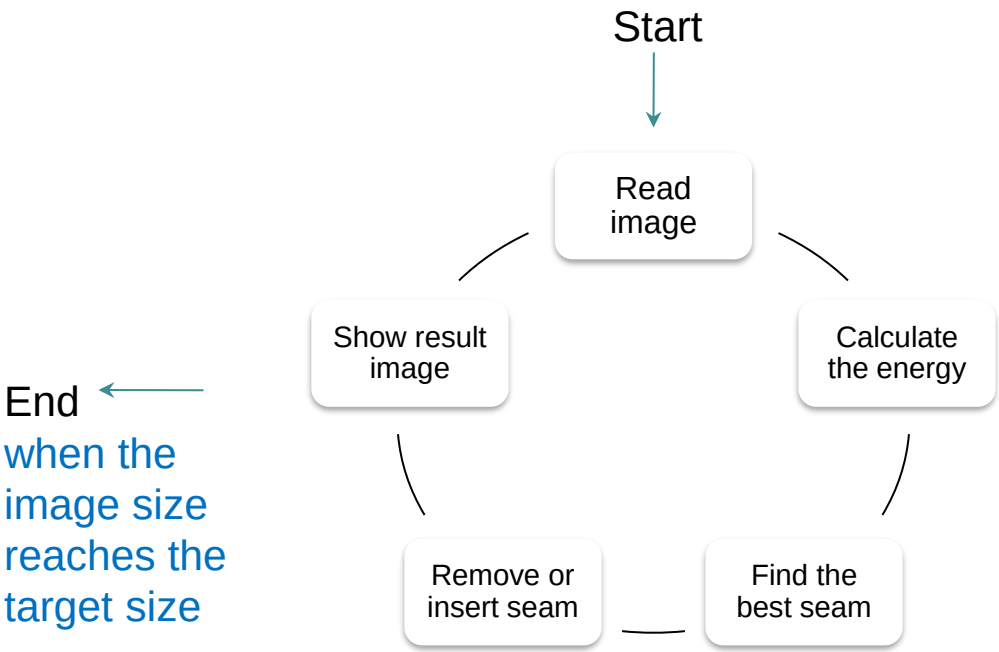


Seam Carving Result



Scaling

Seam Carving Algorithm

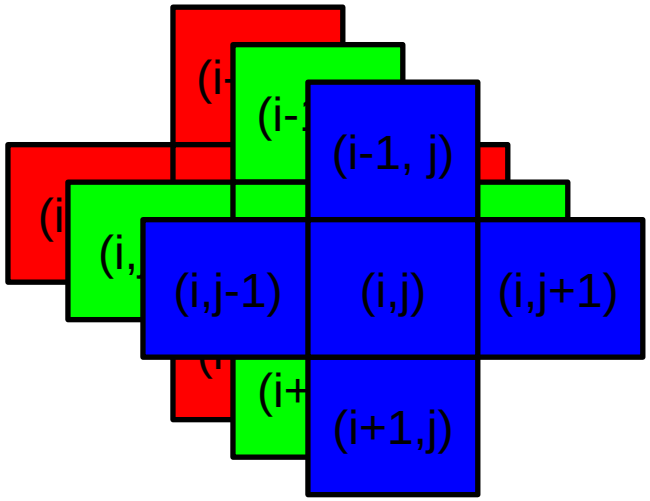


Calculate Energy (Backward)

- Use to calculate the importance:

$$e_1(I) = \left|\frac{\partial}{\partial x} I\right| + \left|\frac{\partial}{\partial y} I\right|$$

- Suppose we want to find the energy at pixel of position (i, j) which calculate for all RGB colors

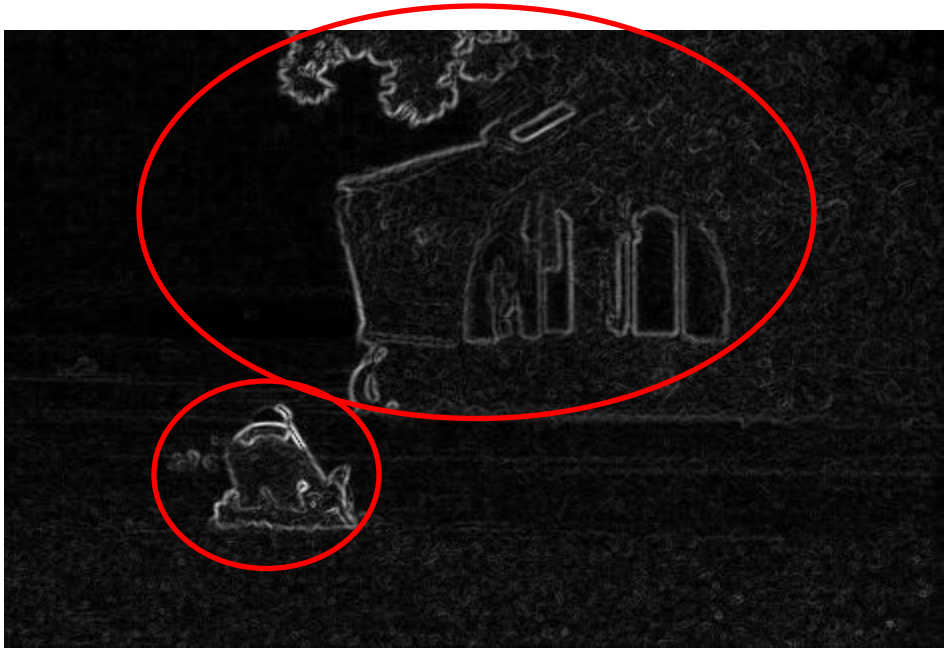


Calculate Energy (Backward) cont.

- White color represent high energy
- Black color represent low energy



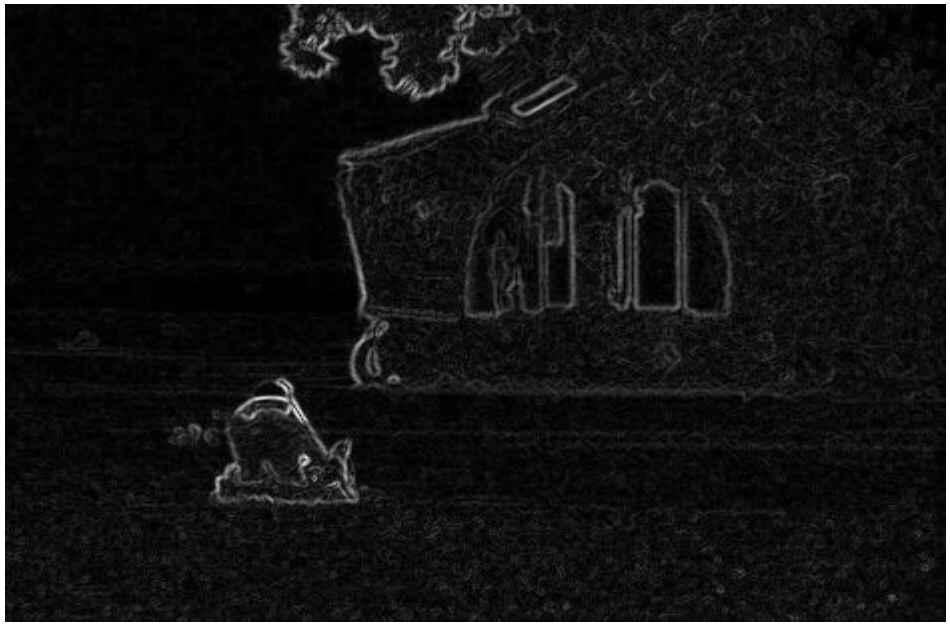
Original Image



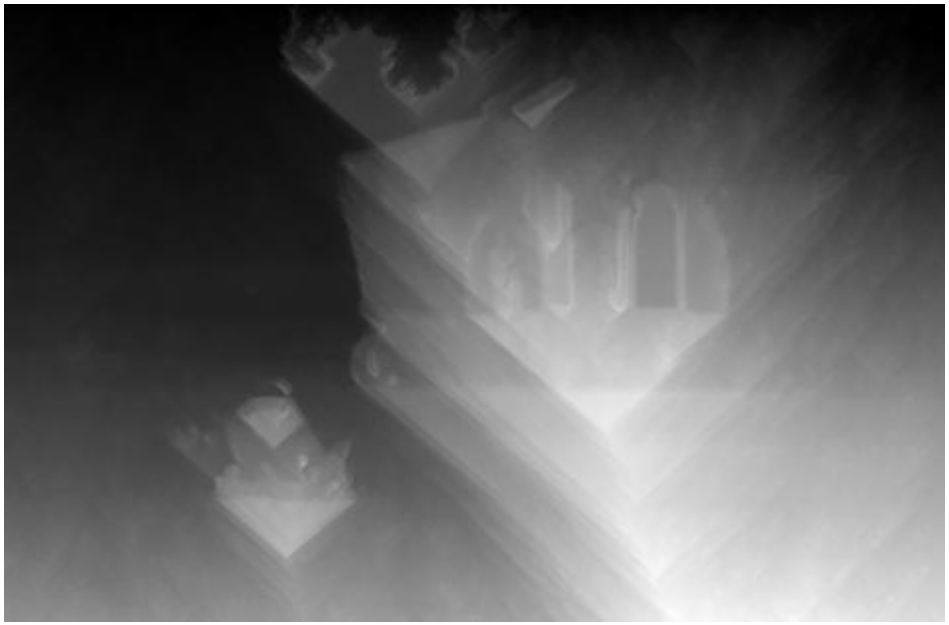
Energy

Find Best Seam (Backward)

- Compute M matrix
  - Use Dynamic Programming
  - M(i, j) = e(i, j) + min(M(i-1, j-1), M(i-1, j), M(i-1, j+1))



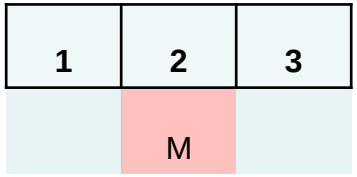
Energy



M Vertical

Find Best Seam (Backward): Dynamic Programing

- Example
  - M(i, j) = e(i, j) + min(M(i-1, j-1), M(i-1, j), M(i-1, j+1))



5.5	16	22.5	13	16
6.5	8	11.5	20	6.5
2.5	9.5	7	7.5	4
5.5	3.5	5.5	10	4.5
6.5	3.5	9.5	9	5.5

Energy

5.5	16	22.5	13	16

M matrix (Backward)

Fill the first row with the energies

0	0	0	0	0

K matrix (Backward)

Find Best Seam (Backward): Dynamic Programing

- Backtrack the seam following the indexes that we keep in K matrix

M matrix (Backward)

5.5	16	22.5	13	16
12	13.5	24.5	33	19.5
14.5	21.5	20.5	27	23.5
20	18	26	30.5	28
24.5	21.5	27.5	35	28.5



Start at the minimum M in the bottom row

0	0	0	0	0
2	1	3	2	1
2	1	1	3	2
2	1	2	1	3
3	2	1	1	2



K matrix (Backward)

Find Best Seam (Backward): Dynamic Programing

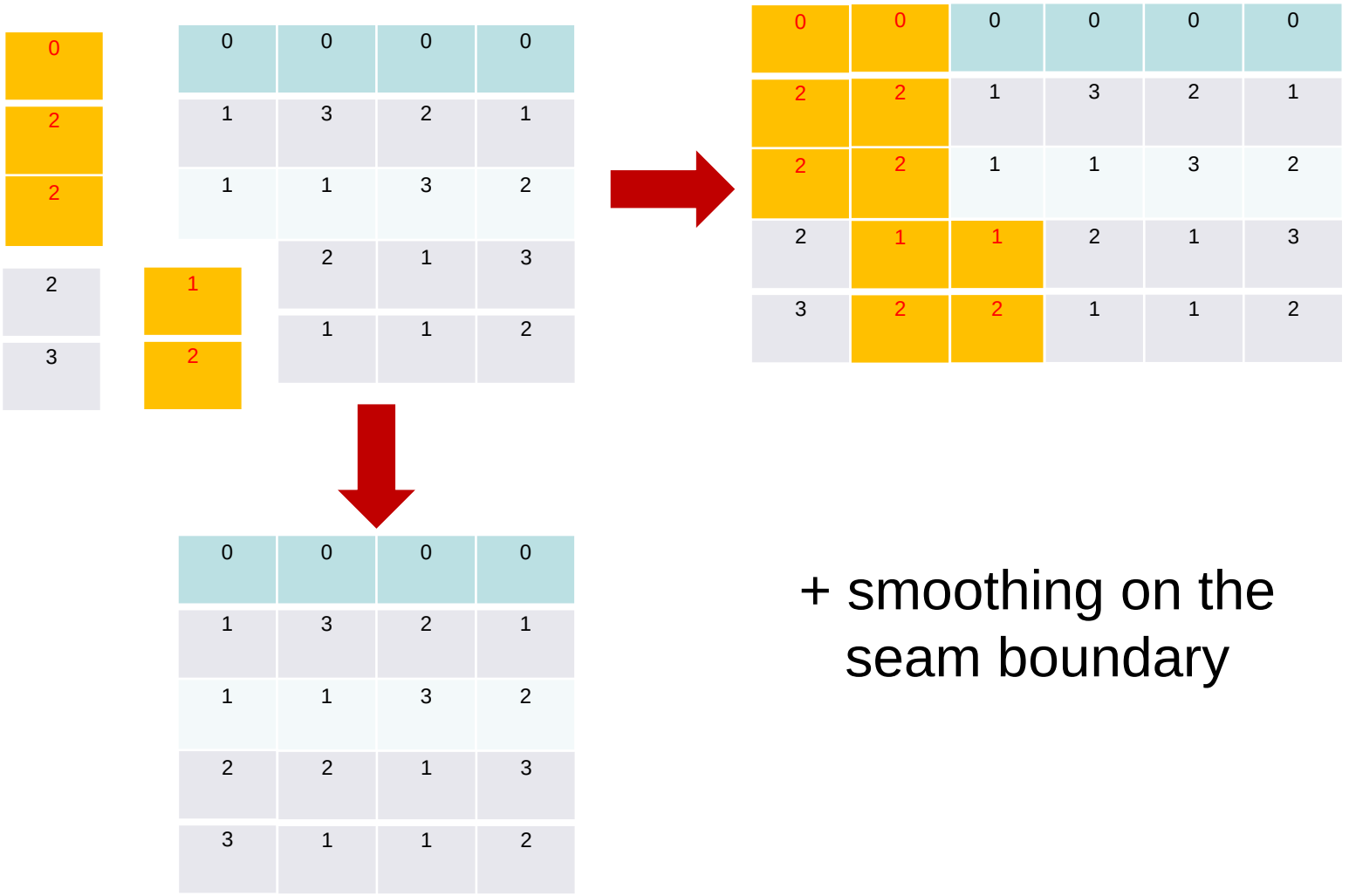
- Backtrack the seam following the indexes that we keep in K matrix

0	0	0	0	0
2	1	3	2	1
2	1	1	3	2
2	1	2	1	3
3	2	1	1	2

K matrix
(Backward)

Find Best Seam (Backward): Dynamic Programing

- Backtrack the seam following the indexes that we keep in K matrix



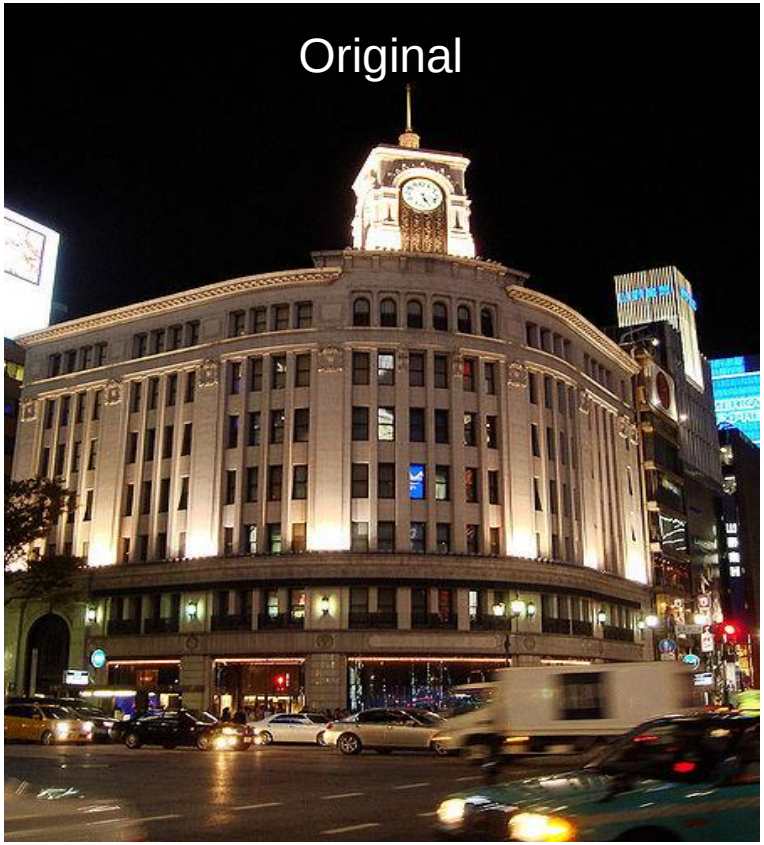
+ smoothing on the seam boundary

Class Activity 2.5

- Based on the seam carving (backward), given the energy image below, find the M and K matrices and indicate the best seam.

4	4	3	13	1
6	8	11	20	6
2	9	7	12	4
5	3	6	1	4
8	2	9	9	23

Backward VS Forward Result

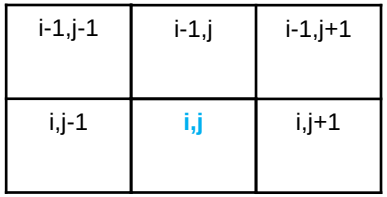


Backward VS Forward Result

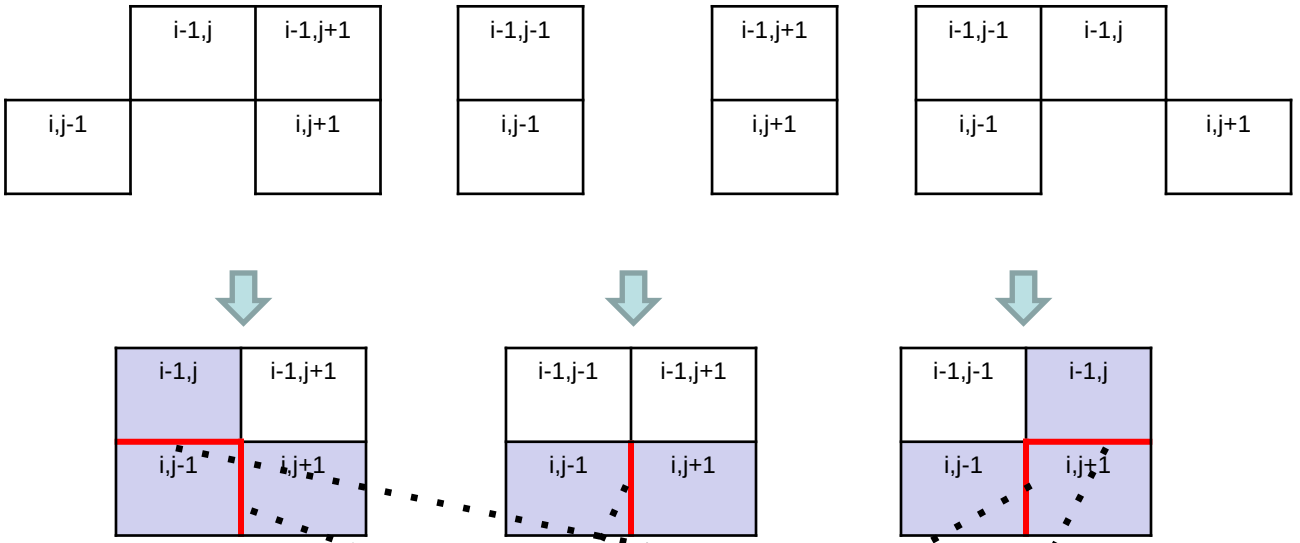


Forward Method will give better result around the edges.





Calculate Energy (Forward)



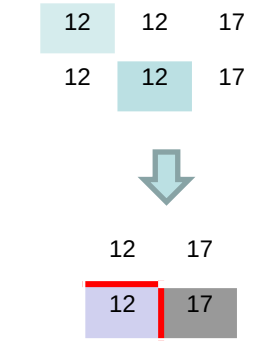
$$C_L(i,j) = |I(i,j+1) - I(i,j-1)| + |I(i-1,j) - I(i,j-1)|$$

$$C_U(i,j) = |I(i,j+1) - I(i,j-1)|$$

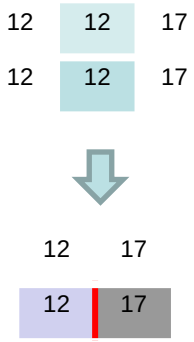
$$C_R(i,j) = |I(i,j+1) - I(i,j-1)| + |I(i-1,j) - I(i,j+1)|$$

Calculate Energy (Forward) cont.

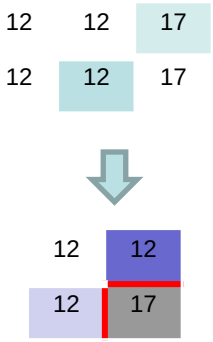
- Example



CL = |17-12| +|12-12| = 5



CU = |17-12| = 5



CR = |17-12| +|12-17|= 10

$$C_L(i,j) = |I(i,j+1) - I(i,j-1)| + |I(i-1,j) - I(i,j-1)|$$

$$C_U(i,j) = |I(i,j+1) - I(i,j-1)|$$

$$C_R(i,j) = |I(i,j+1) - I(i,j-1)| + |I(i-1,j) - I(i,j+1)|$$

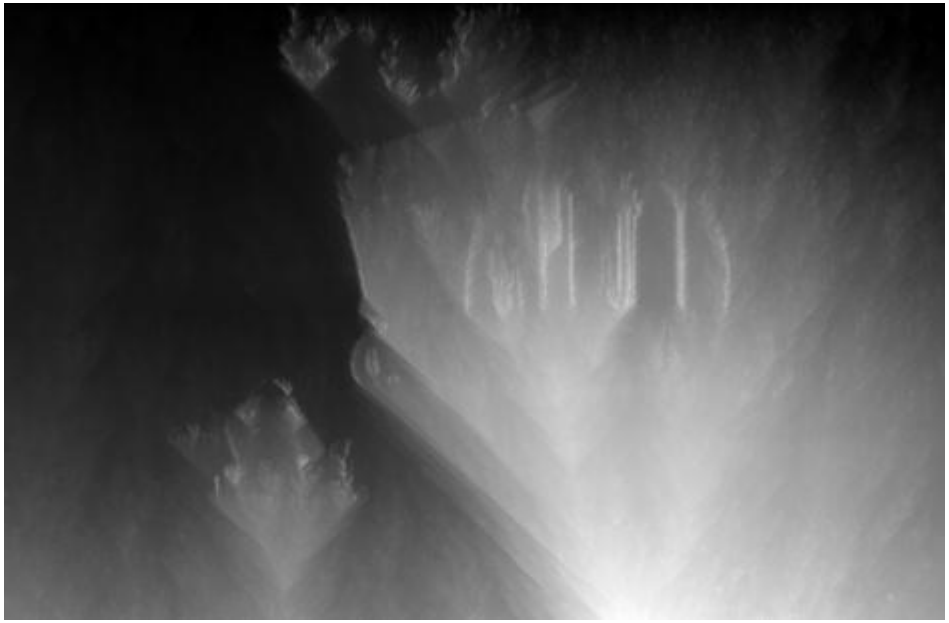
Find Best Seam (Forward)

- Find M matrix

$$M(i,j) = \min \begin{cases} M(i-1, j-1) + C_L(i,j) \\ M(i-1, j) + C_U(i,j) \\ M(i-1, j+1) + C_R(i,j) \end{cases}$$



Original Image



M Vertical

Find Best Seam (Forward) cont.

- Example

$$M(i,j) = \min \begin{cases} M(i-1,j-1) + C_L(i,j) \\ M(i-1,j) + C_U(i,j) \\ M(i-1,j+1) + C_R(i,j) \end{cases}$$

i-1,j-1	i-1,j	i-1,j+1
i,j-1	i,j	i,j+1

$$C_L(i,j) = |I(i,j+1) - I(i,j-1)| + |I(i-1,j) - I(i,j-1)|$$

$$C_U(i,j) = |I(i,j+1) - I(i,j-1)|$$

$$C_R(i,j) = |I(i,j+1) - I(i,j-1)| + |I(i-1,j) - I(i,j+1)|$$

12	17	33	37	19
18	28	8	25	33
15	11	13	22	24
17	11	11	21	27
20	10	14	26	27

Image

5	21	20	14	18

M matrix (Forward)
Fill the first row with CU

0	0	0	0	0

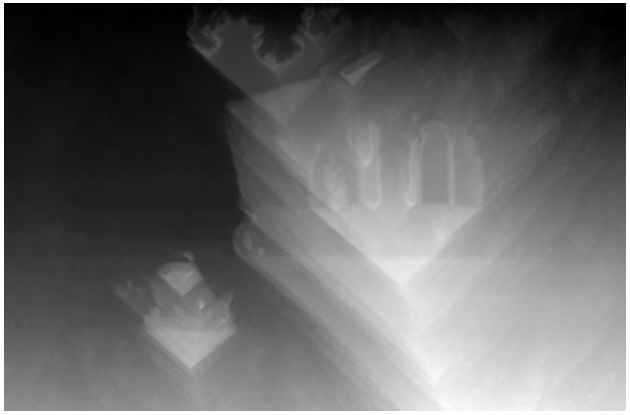
K matrix (Forward)

Find the Best Seam (Forward) cont.

- Differences of M matrix from Backward and Forward Energy



Original Image



M Vertical
(Backward)



M Vertical
(Forward)

Remove or Insert Seam





Remove or Insert Seam



Removal result



Insertion result

Class Activity 2.6

- Based on the seam carving (forward), fill in the values of A and B in the below table (when you want to decrease the **width** of the image).

14	24	35	13	100											
36	18	11	20	60			A	B	C			D	E	F	
21	19	72	120	14											
52	36	46	100	46											
81	12	19	19	23											

Image **M** **K**

- Section1: Find A and D
- Section2: Find B and E
- Section3: Find C and F

Class Activity 2.7

- Based on the seam carving (forward), fill in the values of A and B in the below table (when you want to decrease the **height** of the image).

14	24	35	13	100
36	18	11	20	60
21	19	72	120	14
52	36	46	100	46
81	12	19	19	23

Image

	C			
	B			
	A			

M

	F			
	E			
	D			

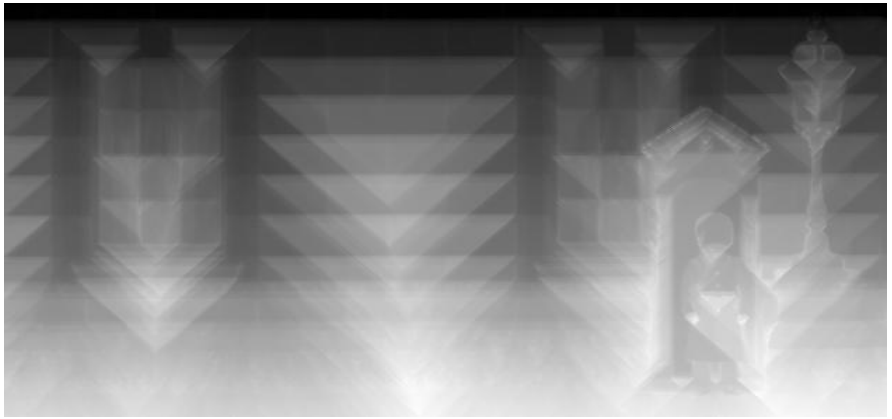
K

- Section1: Find A and D
- Section2: Find B and E
- Section3: Find C and F

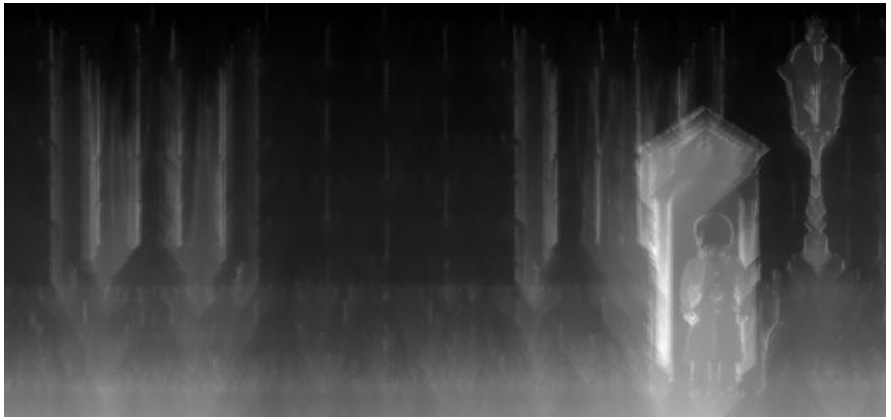
Backward VS Forward Method



Original Image



M matrix - Backward



M matrix - Forward

Backward VS Forward Method



Forward Method will give
better result around the edges.



Backward Result



Forward Result

Seam Carving Applications

- Object Removal



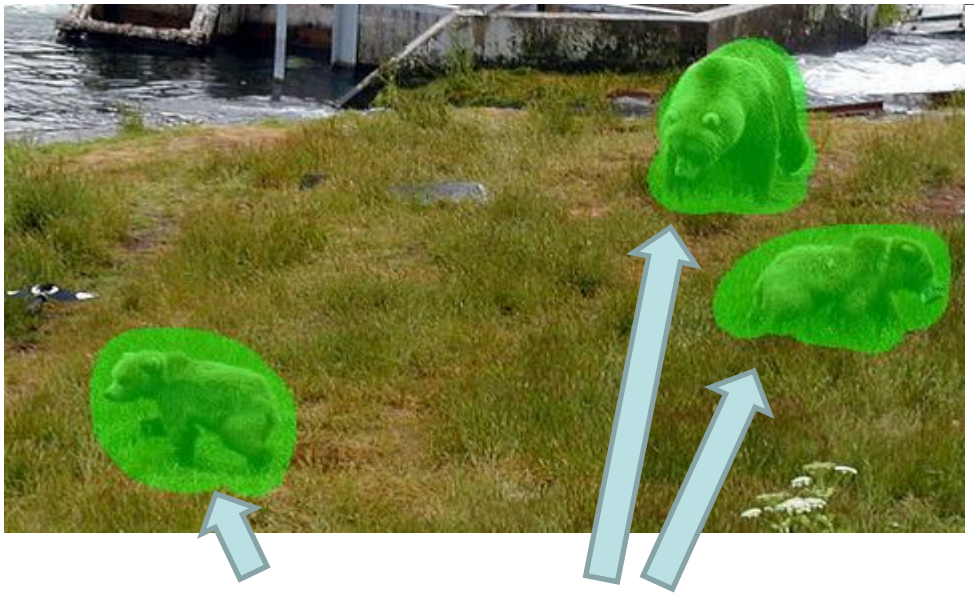
Mask for Removal



Result

Seam Carving Applications (cont.)

- Object Retaining



Mask for Retaining



Result

Class Activity 2.8

- Sample outputs

```
Enter image path (Press Enter to use a sample image):  /kaggle/input/datasets/worapan/mangosteen/mangosteen.jpg

Current Image Size: Width = 390px, Height = 260px
1. Decrease Size (Shrink)
2. Increase Size (Enlarge)
Choose operation (1 or 2):  2

Adjust:
1. Width
2. Height
3. Both
Choose dimension (1, 2, or 3):  1
How many pixels to add to Width (recommended max 195):  100

Processing: Resizing from 390x260 to 490x260...
Tracking vertical seam for expansion 100/100...
Duplicating seams...
Done!
```

Original Image
(390x260)



Seam Carved
(490x260)



Class Activity 2.8

Seam carve can choose

- increase or decrease
- by how many rows/columns smaller/bigger
- show in and out images

Class Activity 2.9

Seam carve can choose

- Modify the code for choosing between
  - Object removal
  - Object retention
- User can identify the location of target object

```
Enter Bounding Box coordinates for target object:
x (min 0, max 500):  100
y (min 0, max 332):  250
width (w):  80
height (h):  80

Select Mode:
1. Object Removal (Carve out the object)
2. Object Retention (Shrink image while protecting the object)
Choose mode (1 or 2):  1

Carving target object across width (80 pixels)...
Carving seam 80/80... Done!
```

Removal

Original + BBox
(500x332)



Target Processed
(420x332)



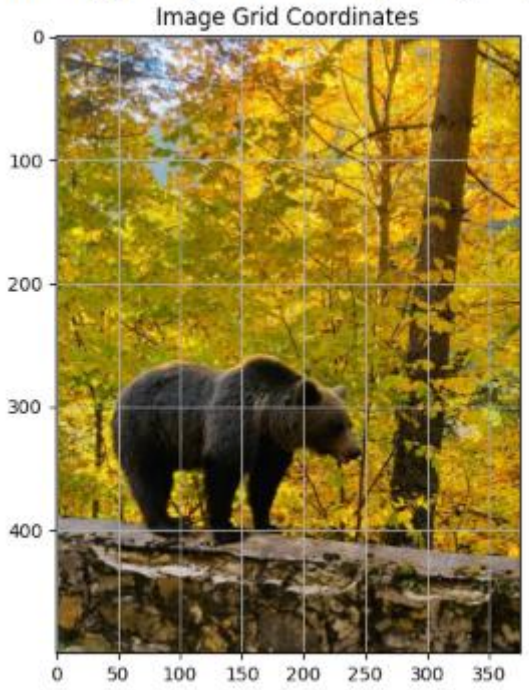
+ Code + Markdown

Class Activity 2.9

Seam carve can choose

- Modify the code for choosing between
  - Object removal
  - Object retention
- User can identify the location of target object

```
Enter image path (Press Enter for sample image):  /kaggle,
```



Retention BW

```
Enter Bounding Box coordinates for target object:
x (min 0, max 375):  40
y (min 0, max 500):  250
width (w):  200
height (h):  150

Select Mode:
1. Object Removal (Carve out the object)
2. Object Retention (Shrink image while protecting the object)
Choose mode (1 or 2):  2

Max safe removal limits: Width <= 165px, Height <= 340px
Pixels to remove from Width (0 to skip):  50
Pixels to remove from Height (0 to skip):  200

Carving 50 vertical seams (Width)...
Carving width seam 50/50... Done!

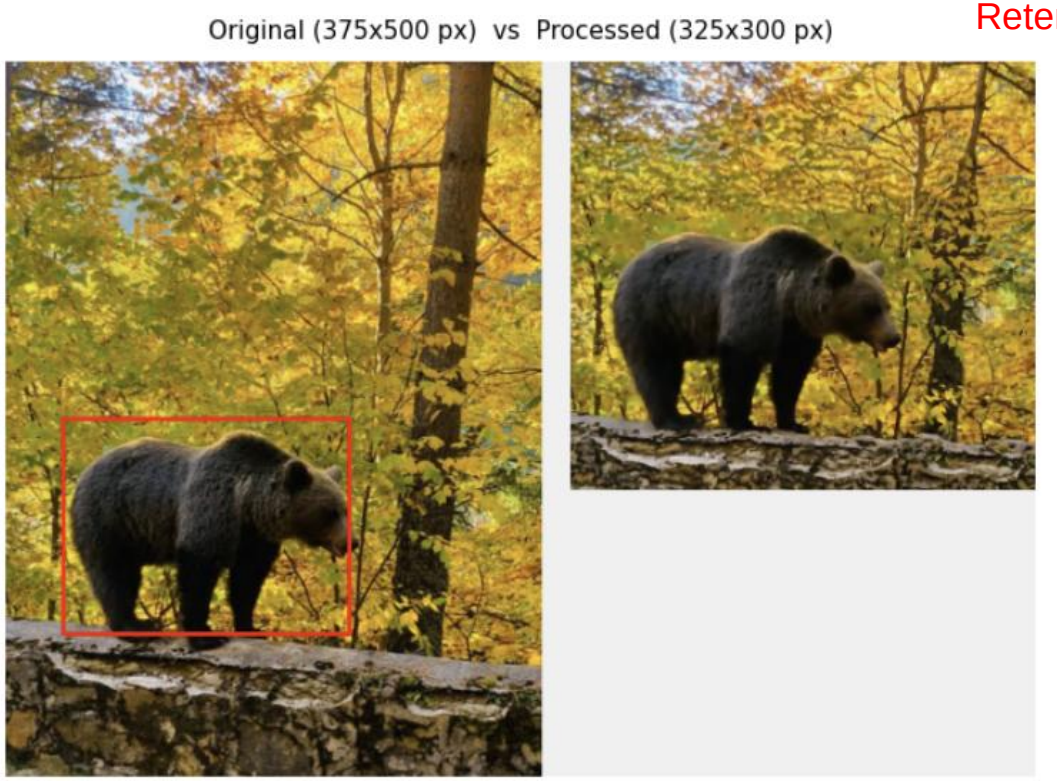
Transposing image & mask for height reduction...
Carving 200 horizontal seams (Height)...
Carving height seam 200/200... Done!
```

Class Activity 2.9

Seam carve can choose

- Modify the code for choosing between
  - Object removal
  - Object retention
- User can identify the location of target object

Retention BW



Class Activity 2.9

Sea

- M
  - –
  - –
- U

```
Enter image path (Press Enter for sample image):  /kaggle/input/datasets/worapan/2026-215-cv-2-seam-remove/seam5.jpg
```




```
Enter Bounding Box coordinates for target object:
x (min 0, max 473):  330
y (min 0, max 221):  50
width (w):  80
height (h):  160

Select Mode:
1. Object Removal (Carve out the object)
2. Object Retention (Shrink image while protecting the object)
Choose mode (1 or 2):  2

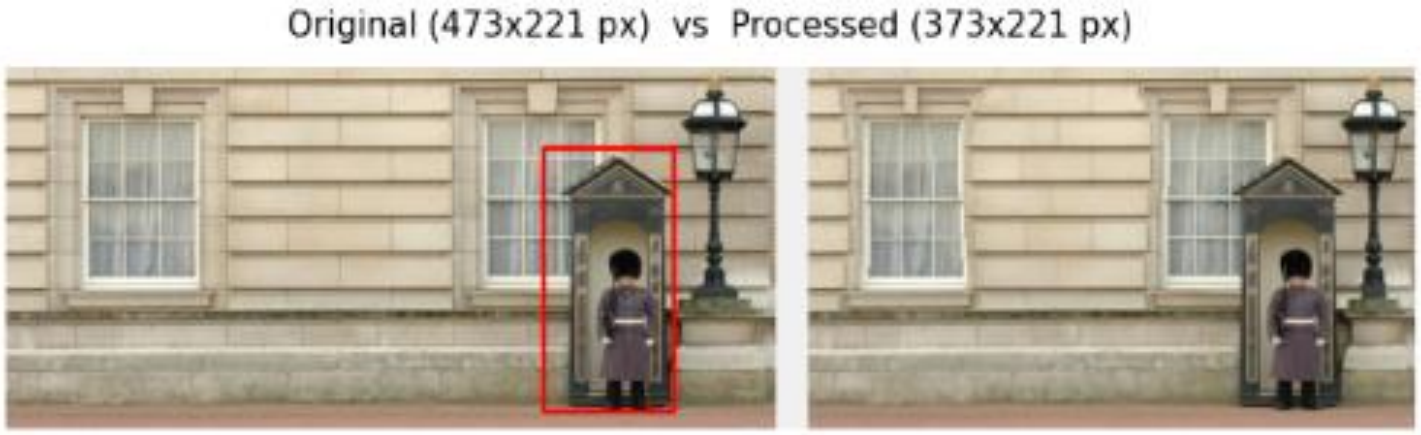
Max safe removal limits: Width <= 383px, Height <= 51px
Pixels to remove from Width (0 to skip):  100
Pixels to remove from Height (0 to skip):  0

Carving 100 vertical seams (Width)...
Carving width seam 100/100... Done!
```

Class Activity 2.9

Seam carve can choose

- Modify the code for choosing between
  - Object removal
  - Object retention
- User can identify the location of target object Retention BW



Class Activity 2.9

Seam carve can choose

- Modify the code for choosing between
  - Object removal
  - Object retention
- User can identify the location of target object

Retention FW



To submit

Individual work (DON'T FORGET to put your name)
Activity 2.1, 2.2, 2.3, 2.5, 2.6, 2.7: Paper-based

Group work (DON'T FORGET to put your members' names)
Activity 2.4
- Use your own picture
- Increase and decrease size for 4 times
- Write a NOTE in comments for
    - Compare between two interpolation methods, which one is better and why
    - Compare between increase and decrease a size, which one has a higher chance to distort an image quality and why

Activity 2.8
Seam carve can choose
- increase or decrease
- by how many rows/columns smaller/bigger
- show in and out images

Activity 2.9
- Object retention / removal